PROJECT DESIGN PHASE-II

SOLUTION REQUIREMENTS SPECIFICATION (SRS)

Date:	29 September 2026	**Project Name:**	**AI BlogNest**
Project ID:	**SWTID-2026-2363**	**Maximum Marks:**	4 Marks

PROJECT OVERVIEW

AI BlogNest is an intelligent content creation and blog management ecosystem. The platform leverages Generative AI to assist authors in drafting, optimizing, and personalizing blog content while offering users personalized content discovery, automated SEO enhancements, and real-time interaction modules.

1. Functional Requirements (FR)

CORE FEATURES & USER STORIES

The functional requirements delineate the essential capabilities, epics, and associated sub-tasks required for end-to-end platform operations.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority
FR-1	**User Registration & Onboarding**	• Registration through custom web form • Single Sign-On (SSO) integration via Gmail • Single Sign-On (SSO) integration via LinkedIn • User persona and interest profile selection upon first login	HIGH
FR-2	**User Account Verification**	• Account verification via automated magic email link • Multi-factor auth confirmation via mobile OTP • Password reset and recovery workflows	HIGH
FR-3	**AI Content Generation Suite**	• AI-assisted blog outline & title generator • Automated blog draft generation based on user prompt/topic • Real-time grammar, tone adjustment, and readability optimizer • Automatic text-to-image generation for blog banners	HIGH
FR-4	**Blog Management & Publishing**	• Rich Text (WYSIWYG) and Markdown editor support • Draft saving, versioning, and post scheduling • Category management, dynamic tagging, and SEO metadata input	HIGH
FR-5	**Recommendation & Analytics**	• Personalized AI feed based on reading history • Author dashboard tracking post views, likes, and reading time • Automated content categorization and key term extraction	MEDIUM

2. Non-Functional Requirements (NFR)

QUALITY ATTRIBUTES & SYSTEM CONSTRAINTS

Non-functional requirements specify the criteria used to judge the operation of the system, rather than specific behaviors.

NFR No.	Quality Attribute	Description & Target Acceptance Criteria
NFR-1	**Usability**	• Clean, modern, and accessible user interface conforming to WCAG 2.1 AA guidelines. • Fully responsive design supporting desktop, tablet, and mobile browsers. • Maximum 3-click navigation path to access core blogging and reading features.
NFR-2	**Security**	• End-to-end data encryption in transit via TLS 1.3 and at rest using AES-256 encryption. • Role-Based Access Control (RBAC) separating Readers, Authors, and System Admins. • Protection against OWASP Top 10 vulnerabilities (SQLi, XSS, CSRF).
NFR-3	**Reliability**	• System uptime SLA of 99.9% excluding scheduled maintenance windows. • Automated database backups every 6 hours with a Recovery Point Objective (RPO) < 1 hour. • Graceful fallback handling if third-party AI APIs experience service downtime.
NFR-4	**Performance**	• Initial page load time under 1.5 seconds under normal network conditions. • AI text generation response streaming initiation within 2.0 seconds of request. • Database query response time under 100ms for read operations.
NFR-5	**Availability**	• 24/7 continuous accessibility via multi-region cloud hosting infrastructure. • Zero-downtime deployment pipeline utilizing blue-green deployment strategies.
NFR-6	**Scalability**	• Horizontal auto-scaling capabilities to handle peak concurrent user spikes up to 10,000 requests/minute. • Stateless microservice architecture for independent component scaling.

3. Validation & Sign-off Criteria

Prepared By: AI BlogNest Project Team **Role:** Solution Architecture & Development **Status:** Approved for Phase-II Implementation	**Reviewed By:** Project Evaluator / Mentor **Signature:** **Date:** / / 2026